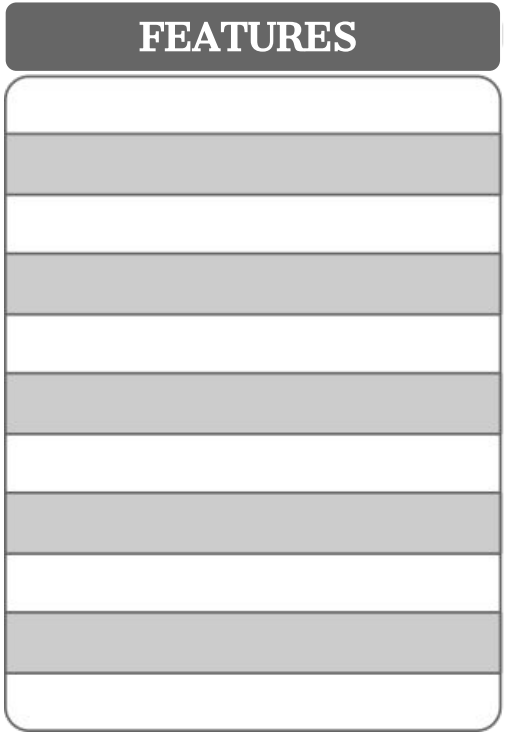
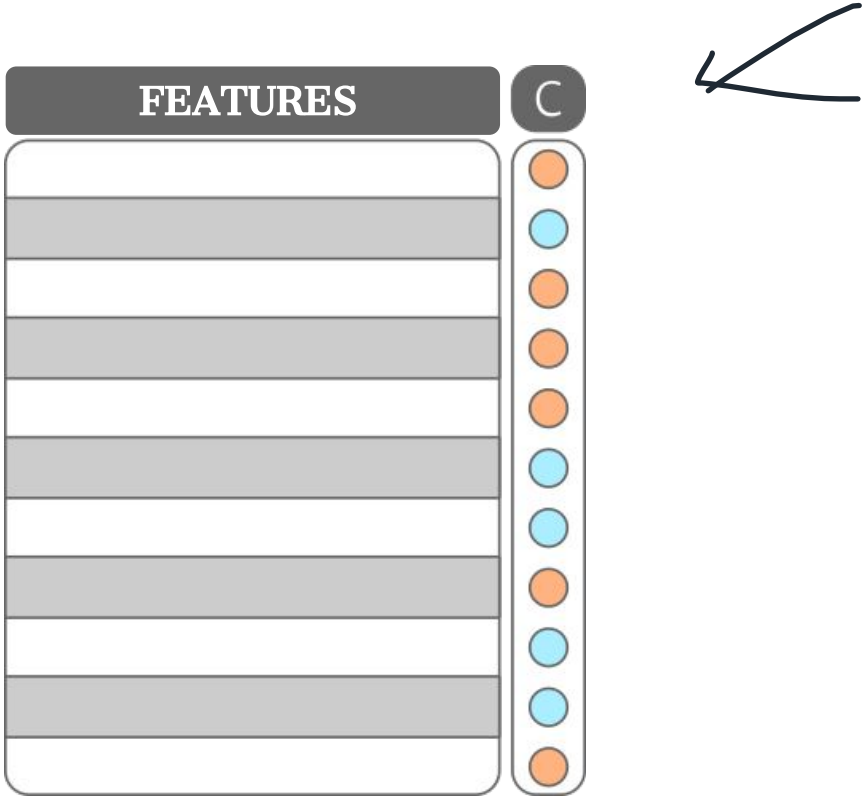
INTRODUCTION TO SPATIO-TEMPORAL DATA

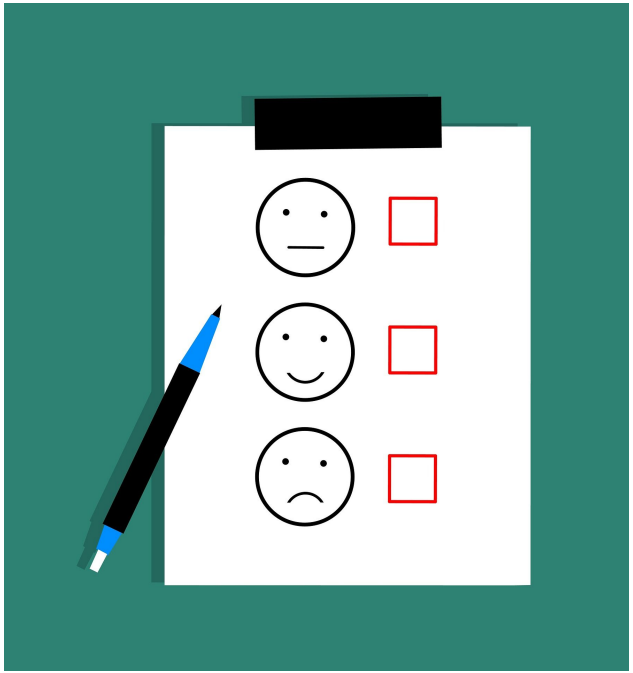
A BIT OF CONTEXT...



A BIT OF CONTEXT...



Examples

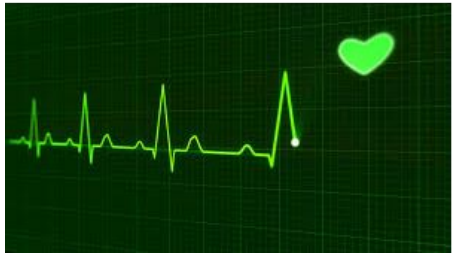


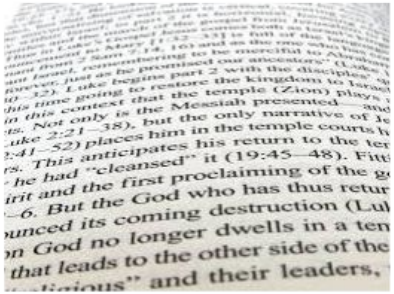
The answers of a person to a survey form a **vector**.

Examples

The characteristics of a movie (gender, duration, country, budget, etc.) form a **vector**.

But...

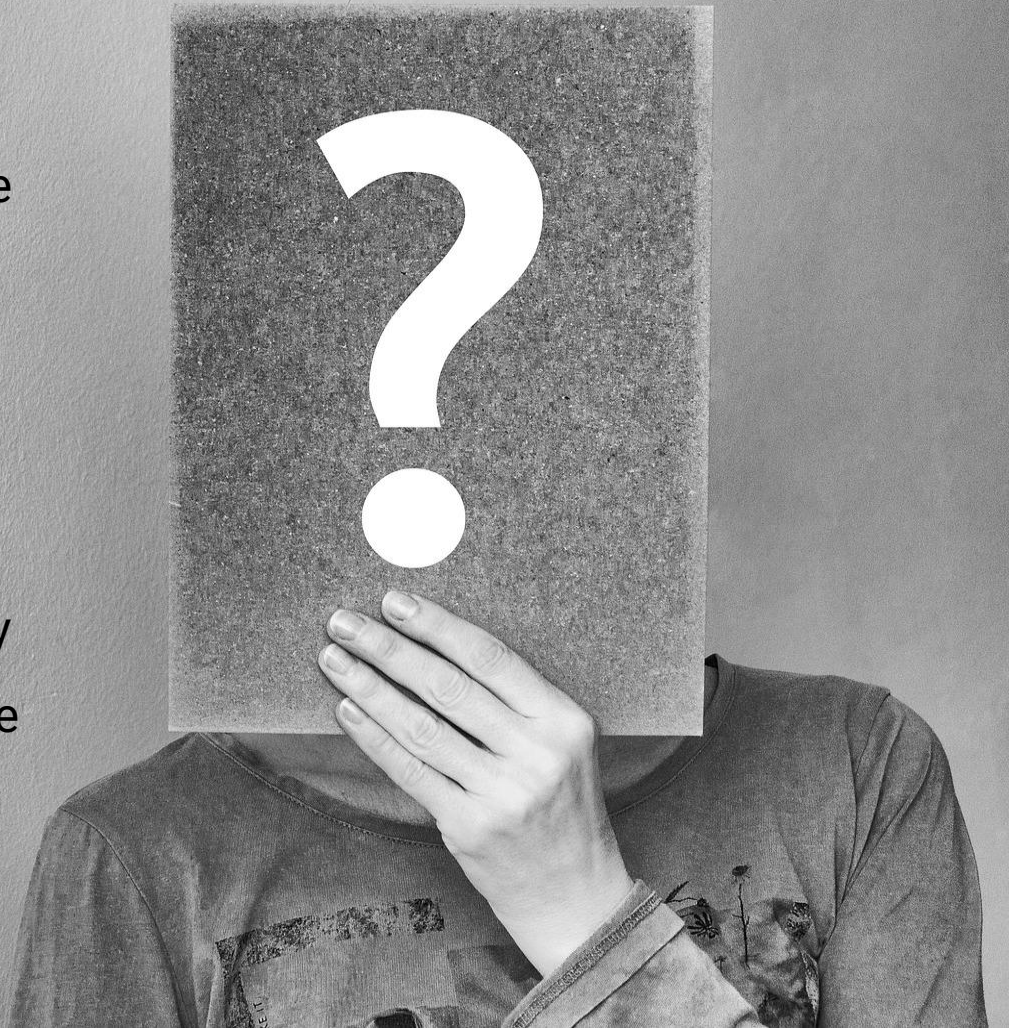




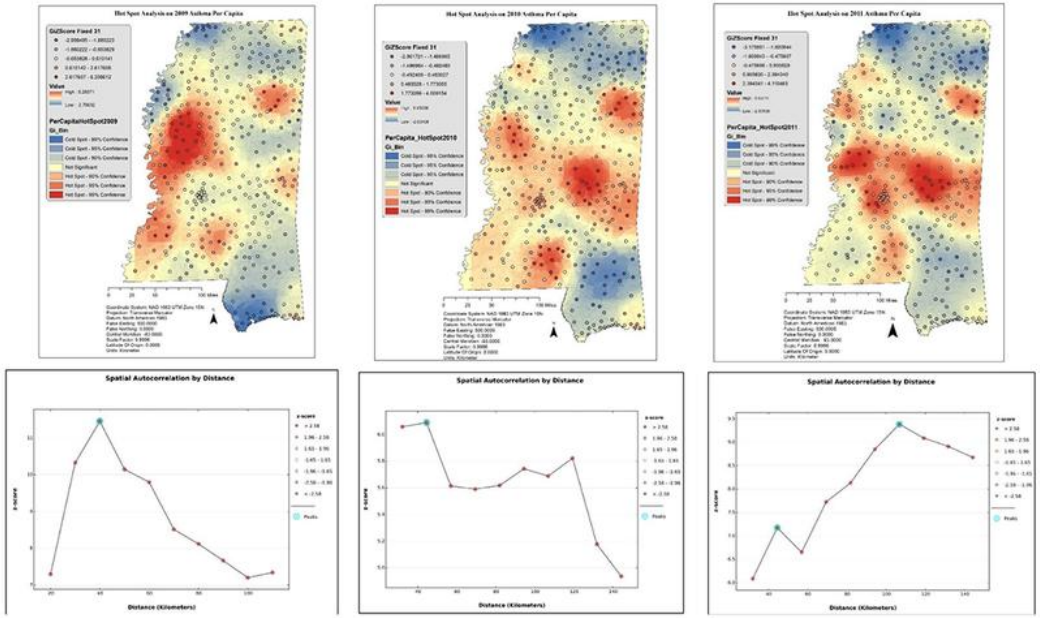
... there are many other types of data that contains extremely useful information ...

- We can represent the data object we have initially by means of a vector and use conventional statistical/ML methods.

- We can design methods specifically designed for the type of data that we have.



Our object of study: spatio-temporal data



2009 2010 2011

Data that represents the **evolution** or one or more attributes (variables) over **space** and **time** (or another ordered dimension).

TOPICS THAT WE WILL STUDY

- **TOPIC 1:** Visualization of spatio-temporal data
- **TOPIC 2:** Time series forecasting
- **TOPIC 3:** Unsupervised classification of time series
- **TOPIC 4:** Supervised classification of time series
- **TOPIC 5:** Analysis of spatio-temporal data

CONTINUOUS ASSESMENT

Written exam: 30%, and a minimum of 4 is required to pass the course.

Project 1 (Dashboard): 20% and an extra 0.5 points for selected project.

Project 2 (Competition): Participation is required, extra 0.5 points for winner.

Project 3 (Poster): 20% and an extra 0.25 points for the best 25% posters.

Project 4 (Tutorial and oral presentation): 30%

CONTINUOUS ASSESMENT

Week	Date	Monday	Tuesday	Friday	Evaluation task deadlines
1	20/01/25 26/01/25			T1: Introduction	
2	27/01/25 02/02/25	P1: Dashboard	P1: Dashboard	P1: Dashboard	
3	03/02/25 09/02/25	P1: Dashboard	P1: Dashboard	P1: Dashboard	P1 - 11/02/24
4	10/02/25 16/02/25	T2: Time series forecasting	Lab 1: TS decomposition	Lab 1 + T2: Time series forecasting	
5	17/02/25 23/02/25	Lab 2: ARIMA models	Lab 2: ARIMA models	T2 + Lab 3: Forecasting with ML	
6	24/02/25 02/03/25	P2: Forecasting competition	Fixed timetable week		P2 - 03/03/24
7	03/03/25 09/03/25	T3: Distance measures for TS	Special Activities	Lab 3: Distance measures for TS	
8	10/03/25 16/03/25	T4: Clustering TS	Lab 4: Clustering TS	T5: Time series classification	
9	17/03/25 23/03/25	P3: TS classification paper	P3: TS classification paper	P3: TS classification paper	
10	24/03/25 30/03/25	Holiday	P3: TS classification paper	P3: Oral presentations	P3 - 01/04/24
11	31/04/25 06/04/25	Special activities	P4: Spatio-temporal data	P4: Spatio-temporal data	
12	07/04/25 13/04/25	P4: Spatio-temporal data	P4: Spatio-temporal data *	Fixed timetable week	** Feedback session*
13	14/04/25 20/04/25	Exam	Holiday	Holiday	
14	28/04/25 04/05/25	P4: Spatio-temporal data	P4: Spatio-temporal data	Special Activities	
15	05/05/25 11/05/25	P4: Spatio-temporal data **	P4: Spatio-temporal data **	P4: Spatio-temporal data **	** Presentations

T: Theory class, Lab: Laboratory class, P: Project class

GLOBAL ASSESMENT

Written exam: 30%, and a minimum of 4 is required to pass the course.

Project 1 (Dashboard): 20% and an extra 0.5 points for selected project.

Project 2 (Tutorial and video presentation): 50%

ARE YOU
READY?

BEFORE ANY TYPE OF MODELING IT IS CRUCIAL TO...

Visualize using different perspectives and types of graphics:

- Calculate main statistics
- Find out about missing data and outliers
- Represent data in different ways

KNOW YOUR DATA!

PROJECT 1

"You work at a small company dedicated to providing innovative data analysis and visualization tools to public and private companies of different domains. ALLSTAT the public company for official statistics is trying out different companies to develop advanced data visualization tools for their data. As a trial, the company has provide a sample of its data, contained in file CO2.xlsx. The objective is to ***develop a prototype visualization Dashboard*** *for this dataset."*

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